IBM Skillsbuild PBL: From Learner to become An AI Agent Architect

Presented By: Gen-Z AI

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Concept Note

Title:

VocalEyes: Hear the world, see the possibilities

Introduction

Many blind and visually impaired people face daily challenges when trying to access information that's easy for others to take for granted — like reading a printed form, understanding a digital image or navigating a document. While some tools exist, they often fall short, especially when it comes to offline use, regional languages or complex visuals like charts and forms.

This project introduces a smart, AI-based system that can read and understand documents or images — then speak the content out loud using natural-sounding speech. It's designed with the needs of blind users in mind: simple, secure, customizable and capable of doing more than just reading — it can explain, describe and truly assist.

Problem Statement:

- For people who are visually impaired, everyday tasks that many of us take for granted can become real challenges. Imagine needing help every time you want to read a form, check the mail, fill out a document or even understand an image shared online. Without the right tools, something as simple as reading a book or even understanding documents shared digitally or navigating a sign becomes a barrier.

- Because of this, many visually impaired individuals have to depend on others just to handle basic things in daily life. It's not just inconvenient — it's a constant reminder of a lack of independence. Everyone deserves the freedom to access information on their own terms, without always needing to ask for help.

Objectives:

- Enable the visually impaired to access any form of visual/textual content.
- Provide seamless, natural language audio outputs from complex documents.
- Design a modular, scalable multi-agent architecture.
- Create a system that is secure, lightweight and deployable in real-life use cases (like education, government, or health services).

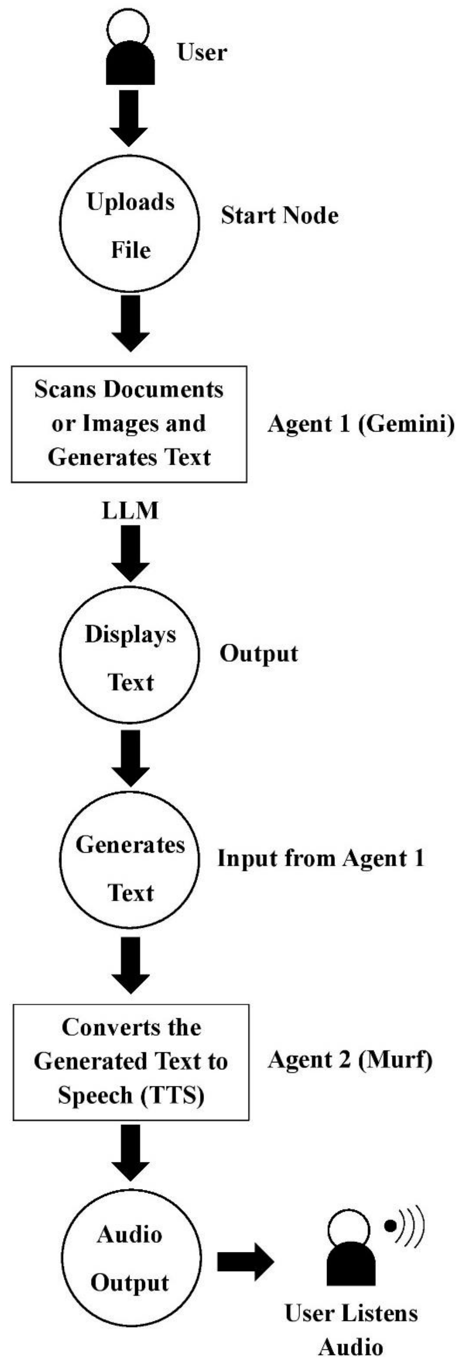
Proposed Solution:

- We propose a Multi-Agent AI System, named **VocalEyes**, to assist blind individuals by converting any uploaded document or image into meaningful audio.
- Agent 1 utilizes the Gemini API to extract and interpret text from images/documents.
- Agent 2 sends the result to Murf API, which converts the content into high-quality speech.
- This end-to-end system allows visually impaired users to upload content and receive accurate, natural-sounding audio output — all powered by intelligent agents.

System Architecture Overview:

1. User uploads a File/Image.
2. Agent 1 (Gemini)
   Extracts content if it is a Document→ Structured text.
   Analyze if it is an Image→ Structured text.
3. Agent 2 (Murf)
   Receives text → Generates Speech.
4. Output: Audio is Streamed or Downloadable.

Workflow of the VocalEyes Multi-Agent System:



Tools Stack:

- Gemini API – Image/document analysis and text extraction.
- Murf API – Text-to-speech synthesis (supports multiple voices and accents).
- Python – For creating and integrating Agents.
- Ipywidgets – For creating interactive UI elements inside Google Colab.
- IPython Display utilities – For rendering outputs like audio and images.
- HTML/CSS – Frontend for file upload.

Technologies Used:

- **OCR (Optical Character Recognition)**

    - OCR is the technology that reads text from images or printed documents. It takes things like scanned pages, photos of signs or handwritten notes and converts them into digital text that a computer can understand.

    - In this project, OCR technology helps extract the text from physical or digital documents so it can be processed and read aloud.

- **TTS (Text-to-Speech)**

    - TTS takes the digital text (from OCR) and turns it into spoken words. It gives the system a voice, allowing blind users to listen to the content instead of reading it.

    - In this project, TTS is what makes the information truly accessible — reading out documents, forms or image content in a clear, natural-sounding voice.

- **Multi-Agent System Architecture**
    - Modular, intelligent agents that handle specific tasks like reading, interpreting, summarizing, or explaining content.
- **AI-Based Image and Document Interpretation**
    - Understands layout, structure, and visual elements (e.g., tables, charts, forms)

Target SDGs:

- **SDG 10: Reduced Inequalities**
    - Helps bridge the information accessibility gap for visually impaired individuals.
    - Enables blind users to consume printed/digital content independently.
- **SDG 4: Quality Education**
    - Supports visually impaired students by enabling them to read learning materials.
    - Potential integration with e-learning platforms or school systems.
- **SDG 3: Good Health and Well-Being**
    - Reduces stress, dependency, and isolation often faced by the blind community.
    - Encourages mental well-being via autonomy.
- **SDG 9: Industry, Innovation and Infrastructure**
    - Uses AI and multi-agent systems in a novel way.
    - Promotes inclusive technology development.

Future Scope:

- Add regional language support.
- Integrate Braille device compatibility.
- Offline support for rural/remote locations.
- Expand to describe images (like charts, graphs, scenes).

Expected Impact:

- Over 285 million visually impaired people globally could benefit from this tool.
- Enhances self-reliance and confidence among blind users.
- Potential integrations with e-Government services, schools, and NGOs.

Conclusion

- This project proposes a multi-agent AI system designed specifically for blind and visually impaired individuals — a tool that can **read documents, interpret images and speak the information out loud**, giving users the power to access content on their own terms.

- But this isn't just another OCR-to-speech tool. It's **modular, secure and customizable**, with the potential to support interpretation of **visual structures** like charts or forms and even explain complex layouts.

- More than just a product, it's a step toward **equal access**, **independent living** and **inclusive innovation** — helping advance global goals like **Reduced Inequalities (SDG 10)**, **Quality Education (SDG 4)** and **Good Health and Well-Being (SDG 3)**.